What Does a Geometric Marker Actually Measure? A Measurement Study of Out-of-Distribution Predictors

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Abstract

Geometric markers of the representation – effective dimension, effective radius, effective utility, participation ratio, numerical rank, and a dozen relatives – are widely used as cheap, training-free proxies for out-of-distribution (OOD) performance, and are reported to outperform in-distribution (ID) test accuracy as a model-selection criterion. We ask a question that the literature has largely skipped: when a geometric marker is said to predict OOD performance, what does it actually measure, and how stable is that measurement? We evaluate twenty-four marker families under one protocol – ID matching around the mean of ID accuracy, then residualization of the architecture main effects – along two axes at once: which implementation produced the numbers, and which distribution shift defines the target. The markers do not behave alike, and neither axis is benign. The effective radius of the manifold flips sign depending on which implementation produced it, from -0.58 on one release to $+0.60$ on ours, and the flip is not confined to the task-shift cell: within a single shift, the two implementations also disagree (jpeg compression: -0.19 versus $+0.20$; pixelate: -0.28 versus $+0.30$). The effective dimension, the one marker that agrees across implementations in sign and magnitude under task shift ($+0.93/+0.92$), reverses sign under photometric shifts, where both implementations agree on the reversal (brightness: -0.49 and -0.71). Across the full map, the sign of the association is a function of the shift type more often than of the marker: eighteen of twenty-four families change sign between the task and noise cells on the one hand and the photometric and compression cells on the other, while the implementations agree with each other far more often than they disagree. A sign reported under one shift does not license a claim under another. A registered fourth cell then asks which part of the implementation axis carries the radius flip: our forty-eight models re-measured under the reference release's host protocol (its feature extraction, sampling seed, downsampling, and participation-ratio formula) with our solver. Every sign survives the swap – the radius stays positive ($+0.602$ own-host, $+0.675$ official-host), so the flip traces to the joint implementation-and-population change rather than to the host protocol – but the host protocol is not a neutral wrapper: it preserves a strong marker's ranking almost exactly (dimension, $\rho = 0.963$) while dragging the two weak markers below the registered consistency threshold ($\rho = 0.877$ and 0.804), a perturbation of the same order as their own signal. We then separate the two reasons a marker can move. Under a degradation sweep, where the amount of information in the input changes, the marker and the linear-probe accuracy fall together, which the information confound and a genuine degradation of the representation both predict. Under a recipe sweep over 42 networks trained on identical inputs, where the information is fixed, the marker tracks nothing: the pooled correlation is negative, and it is carried entirely by the separation between networks that trained successfully and networks that collapsed – inside the high-performing cluster the correlation with accuracy is -0.053 ($p = 0.891$), and removing the recipe effect leaves one marker at $r = -0.079$ ($p = 0.62$). Geometric markers of this family are therefore not measurements of representational quality. We give the full stability map, a four-part recipe for testing the next marker, and the practical rule the map implies: no marker may be recommended

Use of LLMs: Large language models were used as writing and coding assistants in preparing this manuscript; all scientific content, analyses, and reported numbers were determined and verified by the authors.

across shifts, and a single-shift correlation must report its shift as part of the result. The protocol, the analysis code, and all reported numbers are released, and every number in this paper is re-derived from the released data by a script that runs on every commit.

1 Introduction

Representational geometry offers an appealing substitute for a training run. If the geometry of a network's internal representation carried reliable information about how the network would behave under distribution shift, one could select a model without ever running it on the target distribution. Recent work makes exactly this proposal: the effective dimension, effective radius and effective utility of the class-conditioned object manifolds, measured by the GLUE solver, are reported to forecast out-of-distribution performance, and to predict it better than in-distribution test accuracy (Chou et al., 2026).

That claim is now being extended. Geometric quantities – participation ratio, numerical rank, neighborhood smoothness, spectral entropy – are computed on frozen features of models that were never trained for the purpose, and the resulting numbers are used to rank checkpoints, to flag overfitting, or to certify that a representation is "healthy". The measurement is treated as settled: one computes a marker, one correlates it with performance, and if the correlation is strong the marker is a proxy.

This paper is about that step. We take the measurement itself as the object of study, and we ask three questions that a correlation does not answer.

What does the marker measure? When a geometric quantity falls as performance falls, two explanations are available and they are not distinguishable by the observation alone. The representation may have become worse; or the input may have carried less information, in which case any information-bearing quantity must fall as a matter of arithmetic. The second explanation is not exotic: it is what happens whenever the degradation is applied to the input rather than to the training.

Is the measurement stable? A marker is a function of a representation, but in practice it is a function of a marker, an implementation, a set of models, and the shift that defines the target. If the sign of the association depends on which of several implementations produced the numbers, or on which of several shifts was chosen as the target, then the marker is not a property of the representation and cannot be relied on to rank models. This paper varies both axes and reports which of them moves the sign more often.

What should a practitioner do next? A measurement study is only useful if it hands back a procedure. We give a four-part test that a candidate marker has to pass – recipe, implementation, shift, and host protocol – and we identify which of the markers currently in use pass it.

1.1 Contributions

- We measure twenty-four geometric marker families in cells that vary two axes at once – the implementation that produced the numbers and the distribution shift that defines the target – under a single protocol, and we show that they fall into distinct stability classes (Section 4).

- We show that the effective radius of the manifold flips sign across implementations, that the flip also appears within a single shift, and that it is not an artefact of the matching bandwidth. We show that the effective dimension – the one marker that agrees across implementations under task shift – reverses sign under photometric shifts, with both implementations agreeing on the reversal (Section 4).

- We show that the sign of the association is determined by the shift type more often than by the marker: eighteen of twenty-four families change sign between the task/noise cells and the photometric/compression cells, and no marker is stable across all shifts we test (Section 4).

- We separate the information confound from a genuine degradation of the representation using a degradation sweep and a recipe sweep, and we show that once the input information is held fixed the markers carry no information about performance (Section 5).
- We close the implementation–population confound from the protocol side with a registered fourth cell: our models re-measured under the reference release's host protocol. Every sign survives – the radius flip is a joint implementation-and-population effect, not a host protocol effect – while the host protocol emerges as a further degree of freedom that perturbs weak-marker rankings at the order of their own signal (Section 4.3).
- We release the protocol, the analysis code, and a script that re-derives every number in this paper from the released data (Section 8).

2 Related work

Geometric measures of representation quality. Measures of the dimensionality and organisation of a neural representation predate their current use as selection criteria. The GLUE framework (Chung et al., 2018) derives the effective dimension, radius and utility of a set of points from the geometry of the manifold they occupy, via a statistical-mechanical analysis of linear separability; these quantities characterise how much of the input a population of neurons can encode at once. Parallel lines of work have proposed spectral and neighbourhood quantities of similar intent, among them the participation ratio and the numerical rank of the representation (Jiang et al., 2020), and have compared a large family of such measures against observed generalization gaps on deep networks. The generalisation-measure literature is careful about a number of confounds – most famously the requirement that a measure not be a monotone rescaling of the training loss (Jiang et al., 2020) – and the present paper continues in that spirit by treating the confound question at the level of the measurement rather than the correlation.

Diagnostics and failure prediction. A separate literature uses spectral or geometric quantities to detect training failure rather than to rank healthy models. This is a weaker and, we will argue, more defensible use of the same measurements: detecting that a run went wrong does not require the quantity to be a graded measure of quality. We revisit this use in Section 5.

Controlled comparison as a methodological contribution. Comparisons between a proposed predictor and an accuracy baseline can be meaningfully confounded by the very factors that distinguish the models being compared: if the models differ systematically in accuracy, then any quantity correlated with accuracy will appear to work. This paper applies a four-condition control protocol – ID matching, group residualization, dual correlation statistics, and cross-group sign consistency – to the question of what geometric markers measure (Zhou, 2026). We take the protocol as given and apply it across implementations rather than across models.

3 The protocol

We use one protocol throughout, so that the numbers in different cells are comparable. It has four conditions.

C1: ID matching. We restrict to models whose in-distribution accuracy is within $\pm b$ percentage points of the mean, where b is a bandwidth we vary. The reason is that the quantity being predicted is OOD accuracy, and the models in a release differ systematically in ID accuracy; without matching, any marker that correlates with ID accuracy correlates with OOD accuracy for free.

C2: group residualization. Within the matched subset, we regress out the main effects of the architecture family (dummy-variable OLS) from both the marker and the target, and correlate the residuals. Architectures differ systematically in both accuracy and representation geometry, so this step is what separates a within-architecture relationship from a between-architecture one.

C3: dual correlation statistics. We report Pearson r and Spearman ρ together, and we require them to agree in sign. A marker whose two statistics disagree is not usable as a predictor even in the cell where it is measured: the disagreement means a few extreme points are carrying the result.

C4: cross-group sign consistency. We require the within-group association to have the same sign in every group with $n_g \geq 5$.

All numbers reported here are computed with the released code; the script that does so is described in Section 8.

4 The stability map

4.1 Two axes and the cells they define

We measure the same marker in cells that vary two axes at once.

Axis 1: the implementation. Three sources provide marker values.

- **Reference implementation, reference population.** The per-model measurements released with the original study for a 224-model collection spanning five architectures, with the GLUE quantities computed by the reference solver.
- **Reference implementation, second population.** A second release of the same study, covering 80 models measured under six corruption scenarios. The two releases share no models.
- **Own implementation, own population.** Forty-eight ResNet-18 models we trained ourselves, on which we computed the GLUE quantities with our own implementation of the solver (Chung et al., 2018), following the published algorithm. Our implementation is used only for relative comparison within our own family, never against absolute reference values.
- **Official host protocol, own solver, own population.** A fourth cell added to close the implementation–population confound (Section 4.3): the same forty-eight models, measured by our solver but embedded in the reference release's host protocol – the reference feature extraction, the reference preprocessing, the reference sampling and downsampling, and the reference participation-ratio and NC1 formulas. The only difference from the third cell is the host protocol around the solver.

Where the official marker name and our field name differ (for example `dimension` and `glue_dimension`), the two are paired as one family; without this aliasing a cross-implementation pairing does not occur at all.

Axis 2: the distribution shift. Seven shifts define the target variable: the cross-task shift (CIFAR-100 linear probes and Tiny-ImageNet linear probes), and six CIFAR-10-C corruption scenarios (gaussian noise, brightness, contrast, fog, jpeg compression, pixelation) measured as accuracy on the corrupted test set. For the official 80-model release we verified that the corruption scenario changes *only* the corrupted-test accuracy column; all marker columns and the transfer column are bit-identical across the six scenario files. The corruption cells therefore use the corrupted-test accuracy as the target; using the transfer column instead would make all six cells identical by construction.

Each cell of the resulting grid applies the same protocol: ID matching at ± 1 pp around the mean ID accuracy of that cell's models, then architecture residualisation (for our single-architecture population, mean-centering). A family's verdict aggregates its cells by priority: a sign reversal between implementations within a shift is a *flip*; a sign reversal between shifts within an implementation is *shift-dependent*; a sign reversal across matching bandwidths is *unstable* in sample size; agreement in sign everywhere is *stable* (or *weak*, if measured in one source only). Every cell also reports a seeded half-sampling procedure (1,000 draws) that re-estimates the correlation and records how often the sign is retained, so that a verdict is not carried by a single lucky split.

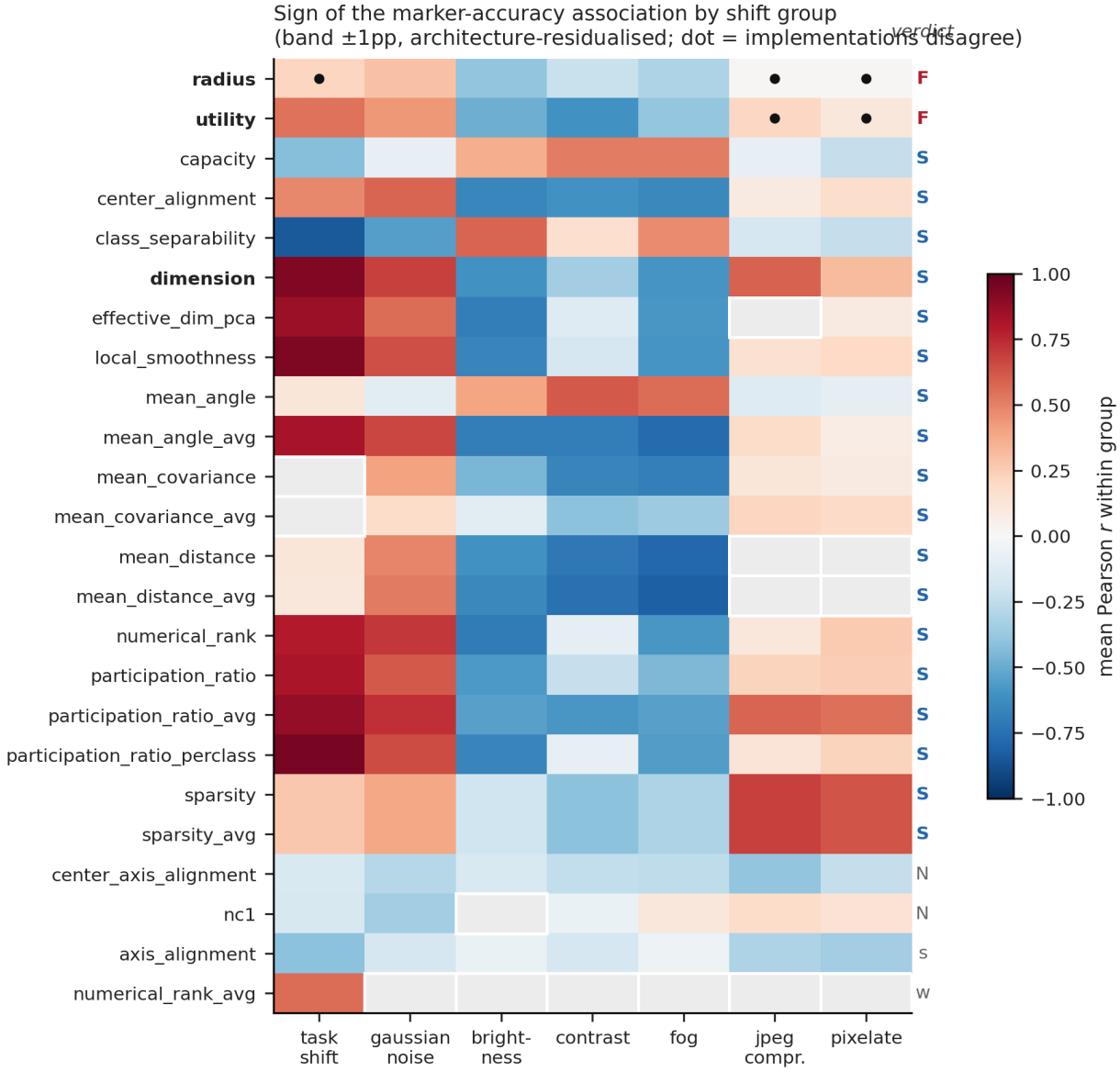


Figure 1: The full sign-stability map. Each row is one marker family; each column is one distribution shift. Colour is the mean Pearson r within that shift (averaged over the implementations that provide the cell); a dot marks a shift where the two implementations disagree in sign; grey means the cell is not measurable at this bandwidth. The right margin letter is the family's verdict (F = flip, S = shift-dependent, N = unstable in sample size, w/s = weak/stable same-sign). Eighteen of twenty-four families are shift-dependent: the photometric and compression columns disagree in sign with the task and noise columns for nearly every marker.

4.2 The map

Figure 1 gives the full map – every family, every shift, both implementations – and Table 1 quotes the cells that carry the argument.

The effective radius flips sign – and not only between implementations. On the reference 224-model population the effective radius is negatively associated with the task-shift target ($r = -0.584$, $n =$

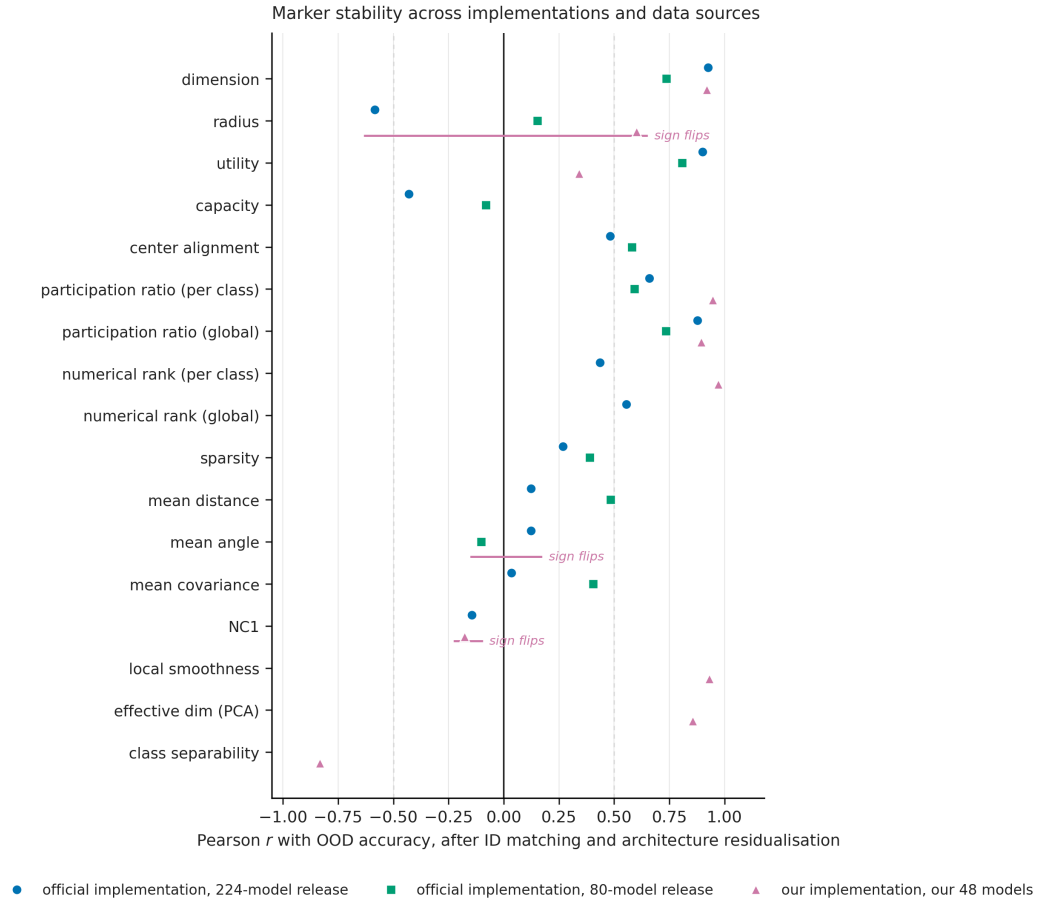


Figure 2: The implementation axis in detail: every marker measured in the three implementation-source cells of the earlier design (reference implementation on the two official releases; our implementation on our own models), under the task-shift target. Each row is one marker; each symbol is one cell; the vertical line is zero. A marker that sits on one side of zero in one cell and the other side in another cannot be used to rank models. This view underlies the per-family discussion below; the full seven-shift view is Figure 1.

106); on our own models it is strongly positive ($r = +0.602$, $n = 32$). The middle cell is near zero (+0.153) and drifts negative as the band widens. This is not a bandwidth artefact: the reference-224 correlation is negative at every bandwidth we tested (±1 to ±5 pp, from −0.584 to −0.796), while our own is positive at every bandwidth (+0.602 to +0.376).

The full map sharpens the finding in a way the three-cell design could not: the implementations also disagree *within* a single shift. Under jpeg compression the reference implementation gives −0.185 and ours +0.204; under pixelation, −0.283 versus +0.301. The flip is therefore not only a property of comparing two model populations across a task boundary – it appears between two implementations measuring two disjoint model sets under a held-fixed corruption type. Where the implementations do agree on the radius (gaussian noise: +0.153 and +0.438), both signs are positive – but the sign there disagrees with the task-shift cells, so the marker fails in both directions: across implementations within a shift, and across shifts within an implementation.

The effective utility flips in the compression cells. The utility agrees in sign between the task cells (+0.901 and +0.342) and collapses in magnitude under our implementation. Under the compression shifts it goes further and flips: jpeg compression +0.542 (reference) versus −0.133 (ours), pixelation +0.411 versus

Family	Shift	ref. impl.	our impl.	n (r/o)	agree	verdict
‘dimension‘	task	+0.925	+0.919	106/32	yes	
‘dimension‘	gaussian noise	+0.737	+0.623	25/32	yes	
‘dimension‘	brightness	−0.493	−0.712	25/32	yes	
‘dimension‘	fog	−0.562	−0.620	25/32	yes	
‘dimension‘	contrast	−0.584	−0.114	25/32	yes	
						shift-dep.
‘radius‘	task	−0.584	+0.602	106/32	**no**	
‘radius‘	jpeg compression	−0.185	+0.204	25/32	**no**	
‘radius‘	pixelate	−0.283	+0.301	25/32	**no**	
‘radius‘	gaussian noise	+0.153	+0.438	25/32	yes	
						flip
‘utility‘	task	+0.901	+0.342	106/32	yes	
‘utility‘	jpeg compression	+0.542	−0.133	25/32	**no**	
‘utility‘	pixelate	+0.411	−0.178	25/32	**no**	
‘utility‘	gaussian noise	+0.808	+0.061	25/32	yes	
						flip
‘participation ratio‘ (global)	task	+0.659	+0.895	106/32	yes	
‘participation ratio‘ (global)	brightness	−0.532	−0.622	25/32	yes	
‘participation ratio‘ (global)	gaussian noise	+0.592	+0.630	25/32	yes	
						shift-dep.

Table 1: The GLUE families and one robust generic quantity, across the shifts that carry the argument, at a matching bandwidth of ±1 pp. Pearson r after ID matching and architecture residualisation; “agree” is sign agreement between the two implementations within the shift; n counts the official/own models entering the cell (own-48 residualisation is mean-centering, since the population is a single architecture). The “verdict” column reproduces the classification assigned by the analysis script and stored in `results/stability_map.json`: *flip* = the implementations disagree within a shift; *shift-dependent* = one implementation reverses sign between shifts. The full map – twenty-four families, seven shifts, Spearman coefficients, sample sizes, bandwidth sweep, and the seeded half-sampling sign-retention rates – is in `results/stability_map.json` and Figure 1.

−0.178. In the earlier three-cell version of this study the utility read as “consistent sign, variable magnitude”; the corruption cells show that reading was an artefact of looking at the task shift only.

The dimension reverses under photometric shifts – with both implementations agreeing. The effective dimension is the one GLUE quantity that agrees in sign and magnitude across implementations under the task shift (+0.925 on 106 models, +0.919 on our 32; also +0.737/+0.623 under gaussian noise). In the earlier version of this study we recommended it on that basis as the one usable marker of the family. The corruption cells overturn that recommendation. Under brightness the dimension correlates negatively with the corrupted-test accuracy under *both* implementations (−0.493 reference, −0.712 ours; our value keeps its sign in 100% of 1,000 seeded half-samples, and the reference value is stable to the bandwidth sweep); under fog, −0.562 and −0.620; under contrast the reference implementation gives −0.584. The sign of the dimension–performance association is a property of the shift type, not of the marker. What survives of the earlier claim is this: the dimension is the only GLUE marker whose sign is consistent *across implementations* within every shift we measured – but no marker, the dimension included, keeps one sign across shifts.

The shift axis dominates the implementation axis. The pattern generalises beyond the GLUE family. Eighteen of the twenty-four families are classified shift-dependent: their sign reverses between the task/noise cells on the one hand and the photometric/compression cells on the other. The numerical rank is the clearest non-GLUE example (+0.972 on our models under the task shift, −0.703 under brightness). Counting cells where both block means are callable ($|r| \geq 0.1$), fourteen families show the block opposition. By contrast the implementations agree with each other far more often than not: among the twenty-one families where the task-shift cell and the gaussian-noise cell are both callable, twenty keep the same sign between those two

cells. The dominant risk to a reported sign is not that someone re-implements the marker; it is that someone applies it to a different shift.

4.3 Closing the confound: the official host protocol on our models

Every implementation-vs-population comparison so far changed two things at once: the solver and the model set. We added a fourth cell that changes one. We ran our solver on our forty-eight models, but embedded it in the host protocol of the reference release: the reference feature-extraction hook (penultimate average-pooled features), the reference per-class sampling (fixed seed, 500 points per class), the reference downsampling, and the reference participation-ratio and NC1 formulas. Nothing about the models changes; nothing about the solver's numerical core changes. The cell was registered with three criteria before it was run: ranking preservation per marker across the two host protocols (Spearman $\rho \geq 0.95$), sign preservation for the radius, and a reference-anchor check on the released example model.

The signs all survive. Under the paper's protocol (ID matching at ± 1 pp, mean-centering, task-shift target, $n = 32$), the dimension reads $+0.919$ in the own-host cell and $+0.924$ in the official-host cell; the radius $+0.602$ versus $+0.675$; the utility $+0.342$ versus $+0.507$. The radius keeps its positive sign under the official host protocol, so the host protocol alone does not produce the radius flip – consistent with the flip arising from the joint change of implementation and population, not from either alone.

The rankings do not all survive, and the pattern of failure is informative. The dimension's ranking is essentially invariant ($\rho = 0.963$, above the registered 0.95 threshold), but the radius ($\rho = 0.877$) and the utility ($\rho = 0.804$) fall below it. The host protocol – sampling seed, downsampling, probe configuration – barely disturbs a strong ranking signal but adds substantial rank noise to weak ones (the radius and utility correlations are ± 0.2–0.4 to begin with). A host protocol is not a neutral wrapper: it is one more degree of freedom in what a marker measures, and for weak markers its perturbation is the same order as the signal.

Two further readings. First, our solver carries a constant systematic offset against the reference values on the released example model (dimension $+32\%$, radius -8%, utility -18%; across ten downsampling seeds the spread is $\pm 4\%$, far below the offset, so it is protocol, not Monte Carlo noise). Because the offset is constant, relative comparisons within a cell are unaffected – which is why the sign agreement above is meaningful despite the offset. Second, the reference participation ratio computed in this cell correlates with the task-shift target at $r = +0.894$ on our models – the same strength class as the dimension – while the reference NC1 does not ($r = -0.244$, Spearman -0.062). The strong markers are strong under both host protocols; the weak ones are not stable under either.

What this licenses. The practical consequence is narrower than the one we drew from the three-cell map, and more useful. Of the three GLUE quantities that the literature reports as interchangeable evidence of a task-relevant manifold, none survives the combined check: the radius and the utility fail across implementations, and the dimension fails across shifts. No marker in the map is stable across all shifts we test. A correlation reported under one shift is a fact about that shift, and any claim of the form "marker X predicts OOD performance" must name the shift in the same breath. The fourth cell sharpens the attribution: the radius flip traces to the joint implementation-and-population change rather than to a host protocol alone, while the host protocol itself emerges as a further degree of freedom – one that leaves strong rankings untouched and perturbs weak markers at the order of their own signal.

5 Two reasons a marker can move

5.1 The confound and the alternative

Suppose a marker falls and performance falls. Two readings are available.

Reading A – the representation degraded. The network learned a worse representation, and the marker registers this.

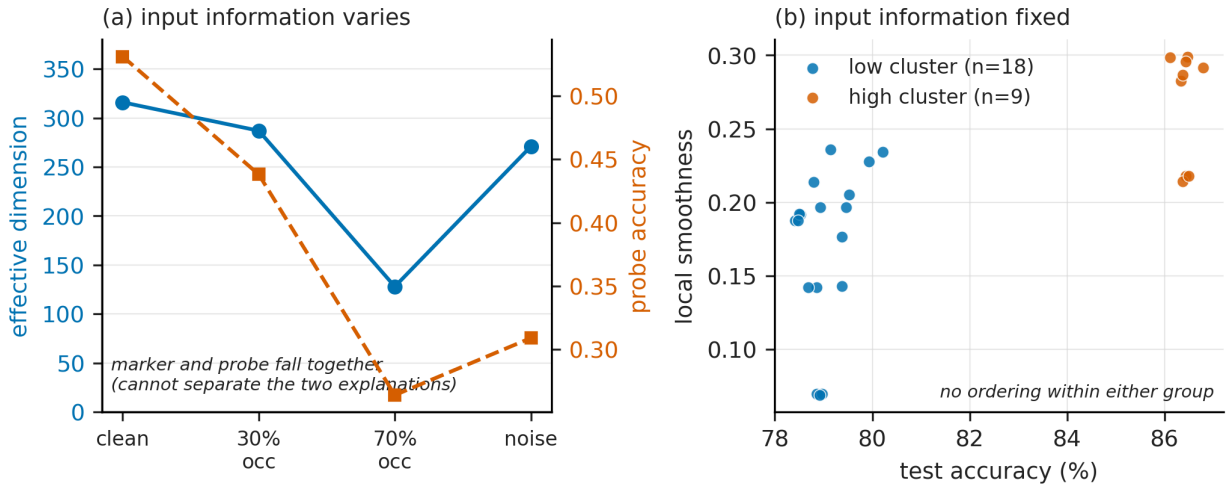


Figure 3: The two comparisons. *Left*: the input information changes, and the marker and the probe accuracy fall together – an observation that both readings predict. *Right*: the input information is fixed and only the training recipe varies, and the marker does not order the models within either performance cluster.

Condition	effective dim	local smoothness	probe acc
clean	316	0.934	0.531
30% occlusion	287	0.910	0.438
70% occlusion	128	0.586	0.264
noise	271	0.899	0.309

Table 2: One network under three degraded inputs. The marker falls by 59% and the probe accuracy by 50% between the clean and the 70%-occluded condition. **Provenance:** these four rows come from a development smoke test (ResNet-50, three seeds, averaged) rather than from the CIFAR-trained networks used elsewhere in this paper, and the per-seed measurements behind them were not retained; they are reported as a diagnostic illustration and are not independently reproducible from the released artifacts. See Limitations 7.

Reading B – the input carried less information. The quantity entering the measurement decreased, and any quantity that measures how much the representation encodes must decrease as a consequence. This is arithmetic, not evidence.

Reading B is not a remote possibility. It is what happens whenever the degradation is applied to the input rather than to the training: occluding a fraction of the image removes information, and the representation of an information-poor input carries less information, by definition.

5.2 Degradation sweep: the input information changes

We take a network trained on clean images and measure it under three degraded inputs: 30% and 70% occlusion, and additive noise. Figure 3 (left) shows the result.

The marker and the probe accuracy move together: the effective dimension falls by 59% and the probe accuracy by 50% between the clean and the heavily occluded condition. This is precisely what Reading B predicts, and it is also what Reading A predicts. **The comparison cannot separate them.** A correlational study that uses occlusion as its degradation would report a strong, significant association and would be unable to tell whether it had measured anything.

One detail is worth recording because it is often read as evidence for Reading A: the marker is not equally sensitive to all degradations. It falls most under 70% occlusion, which removes structure, and least under

Marker	pooled r	p	ρ	within-recipe r
‘effective_dim‘	-0.483	0.0012	-0.536	-0.443
‘local_smoothness‘	-0.452	0.0027	-0.338	-0.079
‘spectral_entropy‘	-0.410	0.0070	-0.185	-0.521
‘criticality_proxy‘	-0.217	0.1678	-0.015	$+0.352$
‘silhouette_gap‘ (trivial)	$+0.768$	$< 10^{-4}$	$+0.732$	—
‘linear_probe_acc‘ (trivial)	$+0.993$	$< 10^{-4}$	$+0.953$	—

Table 3: 42 networks, 14 recipes $\times$ 3 seeds, all trained on identical clean inputs. The two trivial markers use test labels and are a restatement of the target. The within-recipe column removes the recipe mean from both variables, so it asks whether the marker distinguishes networks that differ only in their random seed.

noise, which perturbs amplitude. So it responds to how the information was removed, not only to how much. This is compatible with both readings and does not decide between them.

5.3 Recipe sweep: the input information is fixed

The discriminating experiment holds the input distribution constant and varies only the training. We trained 42 ResNet-18 networks on CIFAR-10 – 14 recipes crossed with 3 seeds – where the recipes span data augmentation, label smoothing, label noise, weight decay, learning rate, epoch budget, and deliberate overfitting. Every network sees exactly the same clean inputs, so the information confound is excluded by construction, and the only thing that differs is the quality of the learned representation.

If a marker measured representational quality, it would order these recipes by quality. It does not.

The pooled correlation is negative and is not a gradient. The pooled correlation of every non-trivial marker with test accuracy is negative (Table 3), the opposite of what a quality measure would give. And the reason is visible in the distribution: test accuracy is sharply bimodal, with networks that trained successfully and networks that collapsed, and nothing in between. The correlation is a separation between the two groups, not a relationship.

Inside a cluster there is nothing. Restricting to the high-performing cluster (three recipes, $n = 9$), the correlation between local smoothness and accuracy is -0.053 ($p = 0.891$). The effective dimension takes the same value for all nine networks, so the correlation is undefined – the marker does not vary at all in the region where the representation is best, and so cannot order it. The direct comparison is stark: the best and the second-best recipe differ in test accuracy by 0.001 and in local smoothness by 0.074. (The accuracy difference is smaller than the seed-to-seed spread of either recipe, so it should not be read as a ranking at all; what the comparison establishes is that the smoothness gap is two orders of magnitude larger than the accuracy gap, not that the top two recipes are distinguishable.)

The marker measures the recipe, not the network. Removing the recipe mean from both variables and correlating the residuals asks whether the marker distinguishes networks that differ only in their random seed. For local smoothness the answer is no: $r = -0.079$ ($p = 0.62$). The marker's value is essentially a property of the recipe.

5.4 One thing the markers are good for

The same measurements separate training failure from training success with accuracy: the criticality proxy reaches an AUC of 0.866 at distinguishing collapsed networks from healthy ones (permutation test over label shuffles, 20,000 draws, $p = 0.003$; 6 failures against 36 healthy runs), the effective dimension and local smoothness reach 0.750 ($p = 0.054$ and $p = 0.053$), and spectral entropy's 0.606 is statistically indistinguishable from chance ($p = 0.43$). Only the first would survive a conventional correction for testing four markers at once, and we report the other three exactly because of that: most geometric markers do *not* separate the two regimes reliably, which is consistent with the reading above. This is a weaker use than grading quality, and it is a defensible one — for the marker that clears the bar. Detecting that a run went wrong does not

require the quantity to be a graded measure of quality; it only requires it to differ between the two regimes. We think this is the right way to use geometric markers today: as a health check on a training run where the specific marker has been validated for that purpose, and not as a measure of the representation.

6 What to do with the next marker

The two comparisons above give a procedure. A candidate geometric marker should be asked two questions, in this order.

1. **Does it survive the recipe sweep?** Train several networks on identical inputs that differ only in the recipe, and require the marker to order them by quality. This is cheap – no new architecture is needed – and it is the test that disqualifies the markers studied here. A marker that moves with the recipe rather than with the network has not been shown to measure anything about the representation.
2. **Does it survive an implementation change?** Compute the marker with two independent implementations, or on two populations that share no models, and require the sign of the association to agree. This is the test that disqualifies the effective radius.
3. **Does it survive a change of shift?** Recompute the correlation with at least one target from a different shift family – a corruption type if the claim was made under task shift, or the reverse – and require the sign to hold. This is the test that the earlier version of this study did not run, and the one that disqualifies the effective dimension. It is also the cheapest of the three: the corruption-cell targets are public, and re-running a correlation costs nothing.
4. **Does it survive a change of host protocol?** Re-measure the same models with the same solver embedded in a different host protocol – different sampling seed, downsampling rule, or probe configuration – and require the ranking to hold (Section 4.3). This is the newest of the four tests and the cheapest to run, since it needs no new models and no new shifts: on our data it left a strong marker's ranking intact ($\rho = 0.963$) while dragging two weak markers below the threshold we had registered ($\rho = 0.877$, 0.804).

A marker that passes all four is worth taking seriously. None of the twenty-four families studied here does: the map is the record of that fact. The most common failure in the current literature is not a wrong sign but an unreported one – a correlation measured under a single shift and read as a property of the marker. Reporting the shift alongside the coefficient, as one reports the dataset alongside an accuracy, would have caught every flip in this paper at the point of writing.

Neither test is expensive, and both are currently skipped. We hope that reporting them as a standard part of any claim about a geometric predictor will make the more confident claims in the literature easier to evaluate.

7 Limitations

1. Our own GLUE implementation follows the published algorithm but does not reproduce two steps of the reference implementation (per-manifold-pair analysis and Gaussianization preprocessing). We therefore use it only for relative comparison within our own model family, never against absolute reference values. A difference between implementations could in principle come from our implementation rather than from the reference one. Two facts bound this concern without removing it: the radius sign also differs between the two official releases, which do not involve our implementation at all; and under brightness and fog the two implementations agree on the sign that refutes the dimension, so the photometric reversal is not an artefact of our solver.
2. The recipe sweep uses one architecture on one dataset. Its finding – that the markers track the recipe rather than the network – is consistent across the three seeds and across the fourteen recipes, but we have not shown that it generalizes to other architectures.

3. The corruption cells of the map use corrupted-test accuracy as the target, not the class-level transfer probe the task cells use; the two targets differ in construction (Section 4). The shift-dependent verdicts therefore compare associations measured against different operationalisations of "performance under shift". We regard this as intrinsic to the question – a shift defines its own target – but a design that held the target fixed and varied only the inputs would separate the two effects.

4. The six corruption scenarios of the official release are correlated with one another (the same 80 models enter every cell), so the six corruption columns are not six independent replications of the photometric/compression pattern. The independent replication of that pattern comes from our own 48 models, which the official release does not contain.

5. We did not have access to the reference implementation of the GLUE solver, so the three-cell design varies the implementation and the population together. We cannot separate the two effects within our own models; a design that holds the population fixed and varies only the implementation would sharpen the claim, and we regard it as the most valuable follow-up. A fourth cell (Section 4.3) narrows this gap from the protocol side: it holds the population and the solver fixed and varies only the host protocol around the solver, showing that the host protocol alone does not reproduce the radius flip but does perturb weak-marker rankings. Full separation – holding everything fixed and swapping only the solver's numerical core – still requires the reference solver's private internals, which remain unavailable; if access is granted the swap becomes a direct check.

6. The degradation sweep is a single network with three seeds, reported as the mean. It is a diagnostic rather than an experiment: it establishes that the degradation sweep cannot discriminate the two readings, not that any particular marker fails. Two further caveats apply to Table 2 specifically. The network is a ResNet-50 measured on a non-CIFAR image set, not one of the CIFAR-trained networks used in the recipe sweep; and the per-seed measurements behind the reported means were not retained, so those four rows cannot be re-derived from anything in the released artifacts. We keep the table because it illustrates the point of the section, but a reader should treat it as an illustration with a stated provenance rather than as a reproducible measurement.

7. We evaluate the markers against one OOD target per cell. The full map multiplies the shifts but not the targets within a shift; whether the stability classes reported here depend on the choice of target within a shift family is an open question.

8 Conclusion

Geometric markers of the representation are useful, and the way to use them well is to stop asking them to grade quality. Our results say five things. First, a marker that falls with performance may be telling you that the input carried less information, and the only way to find out is to hold the input fixed and vary the training. Second, a marker's sign is not guaranteed to survive a change of implementation: the effective radius does not, and the failure repeats within single shifts, not only across model populations. Third, a marker's sign is not guaranteed to survive a change of shift either: the effective dimension – the one marker we had previously recommended – reverses under photometric shifts with both implementations agreeing, and eighteen of twenty-four families in the full map change sign between the task/noise block and the photometric/compression block. Fourth, the implementation–population confound does not reduce to a host-protocol effect: re-measuring our models under the reference host protocol preserves every sign, including the radius's positive sign on our population, while leaving weak-marker rankings perturbed at the order of their own signal – so the flip traces to the joint change, and the host protocol joins the list of degrees of freedom a marker consumer must report. Fifth, the markers are reliable health checks on a training run and unreliable measures of a representation. A practical conclusion follows: report the shift alongside the coefficient, as one reports the dataset alongside an accuracy; run the four tests of Section 6 before believing any new marker; and treat the claims in the literature that name no shift as claims about no particular thing.

Reproducibility statement

All experiments use public data (CIFAR-10/100, CIFAR-10-C, Tiny-ImageNet) and the public releases of the original study (MIT-licensed repository). The protocol implementation, the analysis code, the per-model measurements, and the trained models are released at github.com/zhouranxi/brainai-protocol-a and archived at doi:10.5281/zenodo.23728883 (concept DOI 10.5281/zenodo.23728883); the version DOI of this record is 10.5281/zenodo.23918659.

Every number reported in this paper is re-derived from the released artifacts by the accompanying script `code/audit_paper_measurement.py`, which recomputes every coefficient, sample size and p-value quoted above – including the full sign-stability map of Section 4, read directly from `results/stability_map.json` – and checks each against the value written here. The fourth-cell numbers of Section 4.3 are read from `results/glue_cell4_own48.json` (48 rows) and `results/glue_own48_results.json` against the registered criteria of `code/cell4_verdict.py`; the systematic-offset calibration of our solver is documented by the variant-scan scripts under `code/glue_*`. It requires no GPU and no downloads. The same check runs automatically on every push of the repository, so the reproducibility of this paper is continuously machine-verified rather than asserted.

Ethics statement

This study involves no human participants and no personally identifiable data: all experiments use public benchmark datasets and publicly released per-model measurements from a prior study. The work does not involve clinical, biometric, or otherwise sensitive data, and no individual person is subject to evaluation. We release the protocol implementation, the analysis code, and the trained model checkpoints to permit independent verification and reuse. We are mindful that the models we release are trained on datasets with known licensing terms, and we distribute them subject to those same terms.

Use of AI tools

Large language models were used as writing and coding assistants during preparation of this manuscript: converting an internal draft to LaTeX, assisting with routine refactoring of the analysis scripts, and re-deriving every numeric claim from the released files with an independent verification script. All scientific content — the measurement-study framing, the analyses, the interpretation, and every reported number — was determined by the authors, and each number is checkable against the released data by the accompanying verification script. The authors take full responsibility for the content of the publication.

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